



Assessment 04.3: Unit 4 Test, Spaces We Live In

Unit: Unit 4, Spaces We Live In (Environmental Design and Management) **Given in:** Lesson 4.17 (20 minutes, after the 15-minute review game) **Points:** 50. Topic assessments category. One retake after a review, best score counts. **Covers:** Topic 4.1 Safe, Healthy, Sustainable Homes (items 1 to 5), Topic 4.2 Design: Elements, Principles, Color (items 6 to 11), Topic 4.3 Design a Room project (items 12 to 16), Topic 4.4 Careers (items 17 and 18), then one scenario (item 19) and one short written response (item 20)

Unit 4 Test (student version)

Name: _____ Date: _____ Period: _____

Directions: Answer every item. Write in the space under each one. Show your work on the three math items (4, 11, and 14); a calculator is allowed on those three only. If you are not sure, write what you think and why. You have 20 minutes. The scenario and the written response are at the end; leave five minutes for them.

Part 1: Safe, Healthy, Sustainable Homes

1. (2 points) Housing needs change across a life. Give one thing a home needs when a toddler lives there, and one thing a home needs when an 80-year-old who uses a walker lives there.

Toddler: _____

Walker user: _____

2. (3 points) For each home hazard, write one way to prevent it.

- a) A loose rug at the top of the stairs: _____
- b) A space heater running next to the curtains: _____
- c) Medicines in an unlocked low cabinet in a home with a toddler:

3. (2 points) Name two things every home fire escape plan must include.

1. _____ 2. _____

4. (3 points, show your work) An old bulb in a hallway uses 60 watts. An LED bulb that gives the same light uses 9 watts. The hallway light is on 5 hours a day.

- a) How many watts does the LED save while it is on? _____ watts
- b) How many watt-hours does the LED save in one day? _____ watt-hours
- c) Name one other way, not a light bulb, to conserve energy or water at home.

5. (2 points) Circle the two choices that are green choices (they conserve, reuse, or reduce waste).

- a) Buying a secondhand dresser b) Leaving the lights on when you leave a room c) Installing a low-flow showerhead d) Buying a new plastic bin every time you need storage

Part 2: Design: Elements, Principles, Color

6. (3 points) Write E if the word is an element of design and P if it is a principle of design.

___ color ___ balance ___ texture ___ rhythm ___ line ___ proportion

7. (2 points) What is a focal point in a room? Give one example of something that could be a focal point in a bedroom.

Definition: _____

Example: _____

8. (3 points) Name the color scheme.

a) Blue, blue-green, and green together: _____

b) Red and green together: _____

c) Light blue, medium blue, and navy blue together: _____

9. (2 points) A client wants a small bedroom to feel calm and a little bigger. Should you use mostly warm colors or mostly cool colors? Write one sentence on why.



10. (2 points) On a floor plan, draw the symbol for a door with its swing, and the symbol for a window in a wall.

Door: [space] Window: [space]

11. (3 points, show your work) Scale conversions.

- a) At $1/4$ inch = 1 foot, a wall drawn 3.5 inches long is _____ feet in real life.
- b) At $1/2$ inch = 1 foot, a 12-foot wall is _____ inches on the plan.
- c) At $1/2$ inch = 1 foot on quarter-inch graph paper (two squares per foot), a 36-inch path is _____ squares wide.

Part 3: Design a Room

12. (2 points) A client card says the four things below. Write N for a need (the design fails without it) or W for a want (nice if the money and space allow).

___ Grandpa uses a walker and needs a clear path from the door to the bed. ___ Grandpa would like a TV. ___ The baby's crib must be away from the window and any cords. ___ The couple likes plants.

13. (2 points) Fill in the traffic flow rules from the project.

A path that a walker or wheelchair uses must be at least _____ inches wide. A path anywhere else must be at least _____ inches wide. A wheelchair needs a turning circle _____ inches (_____ feet) across.

14. (3 points, show your work) A Budget Sheet has a subtotal of \$312.00. Sales tax is 8.625 percent. The client's budget is \$400.

- a) The tax is \$_____
- b) The total with tax is \$_____
- c) Is the total under or over the budget, and by how much? _____

15. (2 points) The catalog says a mattress must be bought new, never secondhand. Give one reason. Then name one item you would buy secondhand for a bedroom and one reason it is a smart choice.

Mattress reason: _____

Secondhand item and reason: _____

16. (2 points) For a baby's room, name one green feature and one safety feature you would label on the model, and say why each matters for this client.

Green: _____

Safe: _____

Part 4: Careers

17. (3 points) Match the person to the job. Write the letter.

Jobs: A) electrician B) home inspector C) interior designer D) real estate agent

___ Plans the room's layout, colors, and finishes for a client ___ Wires the new outlet at wheelchair height ___ Checks the finished work against the building code for the buyer

18. (2 points) Name one career from this unit that most people enter through an apprenticeship or trade school after high school, and one that requires a college degree.

Apprenticeship or trade school: _____ College degree: _____

Part 5: Scenario and written response

19. (4 points) A new client. Mr. Boateng, 68, had knee surgery and will use a walker for two months. A first plan for his bedroom shows: the bed against the far wall with its open side facing the window; a 28-inch path from the door to the bed; a thick shag rug in the middle of the floor; a low coffee table between the door and the bed; and the lamp switch on the wall by the door, not by the bed.

List three problems with this plan for this client and the fix for each. Then write the number of inches the path should be.

Problem 1 and fix: _____

Problem 2 and fix: _____

Problem 3 and fix: _____

The path should be _____ inches.

20. (3 points) Your client has \$400. The needs are a bed, a desk, a chair, storage, and a lamp. Priced all new, they add to \$610 before tax. In four to six sentences, explain how you would bring the room under budget without dropping a need. Use the words secondhand, need, want, and quality.











Grade 8 stretch (2 bonus points): if the budget were cut to \$250, what would you change first, and why?



Exit line (on the back, not scored): The item I was surest about was number . ***The one I want to go over is number .***

Teacher key

Points per item are in the item. Accept any answer that is correct and specific even if it is not the one listed. A calculator is allowed on items 4, 11, and 14 only. Grade 6 version: drop the second prompt of item 20 and give the tax on item 14 as “\$26.91” so the student adds. The stretch on item 20 is bonus, 2 points, offered to everyone and required for grade 8 stretch students.

Item	Answer	Points	Standard line checked
1	Toddler: outlet covers, cabinet locks, gates on stairs, cords out of reach, furniture anchored, anything of the kind. Walker user: no stairs or a ramp, wide paths, no loose rugs, grab bars, a bed at the right height, good light, anything of the kind. 1 point each.	2	EDM 1. a); Std 2 PI B
2	a) Remove the rug, or tape or secure it with a non-slip pad; b) move the heater three feet from anything that burns, or turn it off; c) lock the cabinet or move medicines up high and locked. 1 point each; any prevention that would work.	3	EDM 1. b); HSW 3. c)
3	Any two: two ways out of every room; a meeting place outside; smoke alarms on every level, tested; practice the plan; know how to call	2	EDM 1. c); HSW 3. e)

	911; a way to reach a family member who needs help. 1 point each.		
4	a) 60 minus 9 = 51 watts; b) 51 times 5 = 255 watt-hours; c) any conservation method: shorter showers, a low-flow showerhead, cold-water wash, unplugging chargers, a programmable thermostat, sealing drafts, full loads only. 1 point each; work shown on a and b for the point.	3	EDM 1. d); SUS 3. e); CDOS 3a Basic Skills
5	a and c circled. 1 point each; minus 1 if b or d is also circled (minimum 0).	2	EDM 1. d); SUS 2. c)
6	color E, balance P, texture E, rhythm P, line E, proportion P. Half point each.	3	EDM 2. c); National FCS 11.2.1
7	Definition: the first thing the eye goes to in a room; the center of attention (1 point). Example: a headboard, a window, a bold wall, a piece of art, a fireplace (1 point).	2	EDM 2. c); National FCS 11.2.1
8	a) analogous; b) complementary; c) monochromatic. 1 point each.	3	EDM 2. c)
9	Cool colors (blues, greens); they	2	EDM 2. c); National FCS 11.2.2

	recede, feel calm, and make a room feel larger. 1 point for cool, 1 point for a reason.		
10	Door: a gap in the wall with a quarter-circle arc for the swing (1 point). Window: a break in the wall drawn as parallel thin lines (1 point).	2	EDM 2. b); National FCS 11.4.3
11	a) 3.5 divided by 0.25 = 14 feet; b) 12 times 0.5 = 6 inches; c) 36 inches = 3 feet = 6 squares. 1 point each, work shown.	3	EDM 2. b); National FCS 11.4.3; CDOS 3a Basic Skills
12	N, W, N, W. Half point each.	2	EDM 2. a); PSI 2. a) b. and c.
13	36; 30; 60 (5 feet). Half point per blank.	2	EDM 2. b); National FCS 11.4.2
14	a) 312 times 0.08625 = \$26.91; b) 312 plus 26.91 = \$338.91; c) under by \$61.09. 1 point each, work shown. Accept rounding to the cent.	3	EDM 3. a); FCL 4. c); Std 3 PI C; National FCS 11.6.1
15	Mattress: hygiene (bed bugs, dust mites, stains, allergens), or safety for a crib mattress (firmness, fit) (1 point). Secondhand item: a dresser, a desk, a chair, a bookcase, a lamp, a frame, with a reason (solid wood lasts, easy to check for	2	EDM 3. b); FCL 4. f); National FCS 2.1.5

	damage, saves money for a need) (1 point).		
16	Green: an LED lamp, a secondhand dresser, low-VOC paint, a rug of recycled fiber, a window placed for daylight, a plant, with a reason for a baby (air quality, less waste, lower bills) (1 point). Safe: cordless blinds, the crib away from the window and cords, a night light on the path, outlet covers, furniture anchored, nothing in the crib, with a reason (1 point).	2	EDM 2. d); EDM 1. b); SUS 2. f)
17	C, A, B. 1 point each.	3	EDM 4. a); National FCS 11.1.1
18	Apprenticeship or trade: electrician, plumber, HVAC technician, many contractors, furniture maker (1 point). Degree: architect, landscape architect, interior designer (most), sustainability consultant (1 point).	2	EDM 4. b); CCO 2. a); CDOS 1
19	Any three, 1 point each for a problem with a fix that would work: the 28-inch path is too narrow for a walker (widen to 36 inches by	4	EDM 1. b); EDM 2. a), b); PSI 2. a) g. and h.; Std 2 PI B; National FCS 11.4.2

	moving furniture); the shag rug is a trip and catch hazard (remove it, or a flat rug taped down); the coffee table blocks the path (remove it or move it off the path); the lamp switch by the door means walking in the dark (a lamp reachable from the bed, or a switch by the bed, or a night light); the bed's open side faces the window instead of the door (turn the bed so the open side faces the path). Fourth point: 36 inches.		
20	3 points: the answer keeps all five needs (1); it uses secondhand or DIY for some items and says which and why, with a quality check (1); it uses the four words correctly and shows the total coming under \$400 by reasoning, not by dropping a need (1). A strong answer: buy the mattress new, the bed frame and dresser secondhand, the desk DIY or secondhand, the chair secondhand after checking it is	3 (+2)	EDM 3. a), b); FCL 4. c), f); Std 3 PI C; National FCS 2.1.5

	sturdy, and keep the new lamp; the \$210 over becomes under budget. Stretch bonus (2): a clear first change with a reason tied to the client's need (for example, the desk becomes a wall shelf and the bed frame becomes a platform; the mattress stays new).		
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Total: 50 points (plus 2 bonus). Part 1 = 12, Part 2 = 15, Part 3 = 11, Part 4 = 5, Part 5 = 7.

Scoring notes:

- Items 4, 11, and 14 are the math items. Give the point for a correct answer with the work shown; give half a point for correct work with an arithmetic slip.
- Items 19 and 20 are the evaluate-and-justify items. A student who names a problem without a fix on item 19 gets half a point for that line.
- A retake uses the same key with new numbers on items 4, 11, and 14: swap 60 and 9 watts for 75 and 10, the subtotal \$312 for \$276, and 3.5 inches for 2.75 inches.
- ELL: the vocabulary list with translations may be used as a word bank; the written response may be in the home language and scored on content.